

2026/2/25 RFCMP

moonshine現象の紹介

(Introduction to moonshine phenomena)



RFCMP 2026 Mini Workshop

moonshine 現象の紹介 岡田 昌樹

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<https://masakiokada0101.github.io/talks.html> 補足資料

1. Preliminaries from group theory

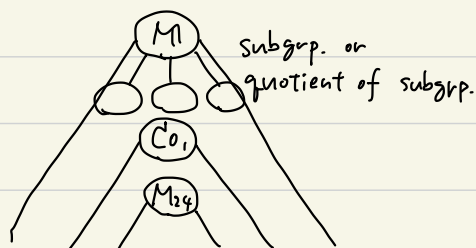
1.1. Finite simple groups Jordan-Hölder theorem

no normal subgrps. other than G and $\{1\}$

Thm (Classification of fin. simple grps.)

A fin. simple grp. is one of the followings.

- cyclic grp. of prime order $\mathbb{Z}_p$ (p : prime)
- alternating grp. A_n ($n \geq 5$)
- simple grps. of Lie type
- 26 sporadic grps.



6 pariahs (not subgrp. quotient of subgrp. of M_1)

○	○
○	○
○	○

Three important fin. simple grps.:

	order
the Mathieu grp. $M_{24} = \text{Aut}(G_{24}) = ?$	2.45×10^8
the Conway grp. $Co_1 = \text{Aut}(L_{24})/\mathbb{Z}_2 = \text{Aut}_{r=1}(V^{\text{fg}})$	4.16×10^{18}
the monster grp. $M = \text{Aut}(B^h) = \text{Aut}(V^h)$ $\parallel$ Griess alg. $V^h _{wt=2}$	8.08×10^{53}

• Conway grps.

Def

A lattice $(L, \langle -, - \rangle)$ of $\overset{L}{\text{dim. rk.}} n$ is a free ab. grp. $L \subset \mathbb{R}^n$ of rk. n whose basis spans $\mathbb{R}^n$ with a non-dgn. sym. bilin form $\langle -, - \rangle : L \times L \rightarrow \mathbb{R}$.

A 24-dim. pos-def. even self-dual lattice is called a Niemeier lattice.

The Leech lattice Λ_{24} is the Niemeier lattice without vectors with $|k|^2 = 2$.
(unique up to isom.)

$$C_{0,0} := \text{Aut}(\Lambda_{24}) \quad \text{Center}(C_{0,0}) = \{\pm 1\} \cong \mathbb{Z}_2$$

$$C_{0,1} := C_{0,0} / \mathbb{Z}_2 \quad \text{simple}$$

• master grp. M

Fact $M = \langle 2^{\overset{\sim}{1+24}} C_{0,1}, \sigma \rangle, \quad \sigma^2 = 1$.
group extension

Def

G is an extraspecial p -group p^{1+n} (p : prime)

$$\Leftrightarrow \begin{cases} \cdot \text{Center}(G) \cong \mathbb{Z}_p \\ \cdot G / \text{Center}(G) \cong (\mathbb{Z}_p)^n \end{cases}$$

ex. gamma matrix algebra

$$\langle \gamma_1, \dots, \gamma_{2m} \mid \{\gamma_i, \gamma_j\} = -2\delta_{ij} \rangle$$

$$\{\text{monomials}\} \cong 2^{1+2m}$$

$$\text{center} = \{\pm 1\}$$

奇数個だと

$\gamma_1 \dots \gamma_{2m+1}$ p^n center に ± 2 まで

1.2. Group extension

how to combine two groups N and G .
generalization of $N \times G$, $N \rtimes G$.

Def

G, N : grps.

An extension of G by N is a short exact seq. of grp. homs.

$$1 \rightarrow N \hookrightarrow \hat{G} \xrightarrow[\text{section } S]{\pi} G \rightarrow 1.$$

($\hat{G}$ is also called an extension. Any extension is denoted by $N.G$.)

- $N \triangleleft \hat{G}$
- $\hat{G}/N \cong G$

$\pi \circ s = \text{id}_G$ not necessarily a grp. hom.

Take a section $S : G \rightarrow \hat{G}$. $N \times G \rightarrow \hat{G}$ is a bijection.

$$\begin{matrix} \downarrow & & \downarrow \\ (n, g) & \mapsto & n \cdot s(g) \end{matrix}$$

- If we can take a grp-hom. section S (If the short ex. seq. splits), then the multiplication law of $\hat{G}$ is $\pi(s(g)n's(g')^{-1}) = g \cdot 1 \cdot g'^{-1} = 1 \therefore s(g)n's(g')^{-1} \in N$

$$\begin{aligned} (n \cdot s(g)) \cdot (n' \cdot s(g')) &= (n \cdot s(g)n's(g')^{-1}) \cdot s(gg') \\ &=: (n \cdot \varphi_g(n')) \cdot s(gg') \end{aligned}$$

where $\varphi_g : N \rightarrow N$ defines a G -action $G \curvearrowright N$.

$\therefore \hat{G} \cong N \rtimes_{\varphi} G$. $N:G$

In particular, if φ is a trivial action $\varphi_g = \text{id}_N$, then $\hat{G} \cong N \times G$.

• Suppose S is not a grp. hom.

We can define $\varepsilon : G \times G \rightarrow N$ s.t.

$$S(g) S(g') = \varepsilon(g, g') S(gg') \quad \pi(\underbrace{S(g) S(g') S(gg')^{-1}}_N) = 1$$

Let N be abelian. Then $\varphi_g : N \rightarrow N$ defines a G -action $G \curvearrowright N$.

$$\begin{aligned} \varphi_g : N &\rightarrow N \\ n &\mapsto S(g) n S(g)^{-1} \end{aligned} \quad \begin{aligned} S(g) S(g') n S(g')^{-1} S(g)^{-1} \\ &= S(gg') n S(gg')^{-1} \\ &\uparrow \\ S(gg')^{-1} S(g') S(g) n &\in N : \text{abelian} \end{aligned}$$

The multi. law of $\hat{G}$ is

$$(n \cdot S(g)) \cdot (n' \cdot S(g')) = (n \cdot \varphi_g(n') \cdot \varepsilon(g, g')) \cdot S(gg')$$

ε is a 2-cocycle :

$$\varepsilon(g, g') \cdot \varepsilon(gg', g'') = \varphi_g(\varepsilon(g', g'')) \varepsilon(g, g'g'') \quad \leftarrow \text{associativity of } \hat{G}$$

Prop

G : a grp.

N : a G -module (an ab. grp. with G -action)

$$\left\{ \begin{array}{l} \text{extensions of } G \text{ by } N \\ \text{compatible with the } G\text{-action} \end{array} \right\} / \text{equiv.} \xrightarrow{1:1} H^2(G; N)$$

$$\begin{aligned} &\left(\begin{array}{l} \text{A set } N \times G \text{ with multi. law} \\ (n, g) \cdot (n', g') = (n \cdot \varphi_g(n') \cdot \varepsilon(g, g'), gg') \end{array} \right) \longleftarrow [\varepsilon] \\ &N \times G \longleftarrow 0 \end{aligned}$$

* Even if $\hat{G} \cong \hat{G}'$, they can be inequiv. as grp. extensions.

$$\begin{array}{ccccc} & & \hat{G} & & \\ 1 & \rightarrow & N & \xrightarrow{\text{inj}} & \hat{G} & \rightarrow & G & \rightarrow 1 \\ & & \downarrow & & \downarrow & & \\ & & N & \xrightarrow{\text{inj}} & G & & \end{array}$$

In particular, if $N \subset \text{Center}(\hat{G})$, then φ is a trivial action.

$$(n \cdot S(g)) \cdot (n' \cdot S(g')) = (n \cdot n' \cdot \varepsilon(g, g')) \cdot S(gg')$$

$\hat{G}$ is called a central extension.

Examples

· Fact $H^2(C_0; \mathbb{Z}_2) \cong \mathbb{Z}_2$

trivial action

$\leadsto$ There are only two central extensions: $\mathbb{Z}_2 \times C_0$ (split) and C_0 (non-split)

· carry (繰り上がり) central extension, non-split

$$\mathbb{Z}_{100} \cong \mathbb{Z}_{10} \cdot \mathbb{Z}_{10}$$

$\uparrow$ $\uparrow$ $\uparrow$
0-9位 +1位 -1位

$$27+38=65$$

$$(2, 7) + (3, 8) = (2+3+\underbrace{2(7 \cdot 8)}_{=1 \text{ (carry)}}, 7+8) = (6, 5)$$

2. Monstrous moonshine

modular j -function · modular inv.
(a mod. func. of weight 0)

$$j_1(\tau) = q^{-1} + 196884q + 21493760q^2 + \dots \quad (q = e^{2\pi i \tau})$$

$$j(\tau) = q^{-1} + 744 + \dots$$

$$\underbrace{1 + 196883}_{1 + 196883} \quad \underbrace{1 + 196883 + 21296876}_{1 + 196883 + 21296876}$$

McKay (1978), Thompson (1979)

dim. of irrep. of M

$$d_1 = 1, d_2 = 196883, d_3 = 21296876, \dots \quad \text{"monstrous moonshine"} \quad \text{Conway Norton}$$

Moreover, if we decompose $j_1(\tau)$ into the irr. characters

$$ch_h(\tau) := \text{Tr}_{V_h} q^{L_0 - \frac{c}{24}} \quad (V_h: \text{the irr. highest-weight rep. of weight } h)$$

of the Virasoro alg. of $c=24$,

$$j_1(\tau) = d_1 ch_0 + d_2 ch_2 + d_3 ch_3 + d_4 ch_4$$

$$+ d_6 ch_5 + (d_5 + d_7) ch_6 + \dots$$

very simple!

This observation was explained by a CFT_{VOA} with M symmetry.

idea A chiral CFT_{VOA} V is a rep. sp. of the Vir. alg.:

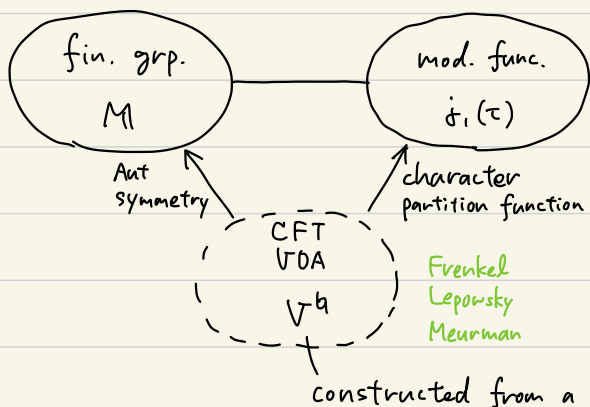
$$V = V_{h_1} \oplus \dots \oplus V_{h_1} \oplus V_{h_2} \oplus \dots \oplus V_{h_2} \oplus \dots$$

If V has the symmetry M ($\exists$ action $M \curvearrowright V$ which commutes with the Vir. action)
($\text{Aut}(V) \supset M$),

$$V = W_1 \otimes V_{h_1} \oplus W_2 \otimes V_{h_2} \oplus \dots$$

Its character is ↖ rep. sp. of M

$$\text{Tr}_V q^{L_0 - \frac{c}{24}} = \sum_i (\dim W_i) ch_{h_i}(\tau)$$



Many other relations

fin. grp.

mod. func. /
weak Jacobi form

are observed.

• lattice VOA

pos-def. even self-dual lattice L of rk. n

$\leadsto$ bosonic mod-inv. VOA V_L of $C = n$

As a vec. sp., $V_L = \text{Span}_{\mathbb{C}} \{ \underbrace{\alpha_{-m_1}^{i_1} \cdots \alpha_{-m_\ell}^{i_\ell}}_{\substack{\text{weight} \\ (L_0\text{-eigenvalue})}} |k\rangle \mid \substack{1 \leq i_j \leq n, \ell = 0, 1, \dots, \\ m_j \in \mathbb{Z}_{>0}, k \in L} \}$

creation operator $[\alpha_m, \alpha_n^\dagger] = m \delta_{m+n,0}$

$m_1 + \cdots + m_\ell + \frac{1}{2}|k|^2$

Its character is

$$\begin{aligned} \text{Tr}_{V_L} q^{L_0 - \frac{c}{24}} &= \frac{\Theta_L(\tau)}{\eta(\tau)^n} \\ &\stackrel{n=24}{=} \cancel{j_1(\tau)} + 24 + \# \{ k \in L \mid |k|^2 = 2 \} \end{aligned}$$

~~$\alpha_{-1}^{i_1} |0\rangle$~~ ~~$|k\rangle$~~ $L_0 - \frac{c}{24} = 0$

Orbifold by $\mathbb{Z}_2$ $L = \Lambda_{24}$
($X^i \mapsto -X^i$)

So, we consider $V^h := V_{\Lambda_{24}} / \mathbb{Z}_2$.

monster VOA (moonshine module)

How $\text{Aut}(V^h) = M$?

$$\Lambda_{24}$$

cocycle factor $\left\{ \begin{array}{l} \varepsilon: L \times L \rightarrow \mathbb{Z}_2 \end{array} \right\}$

central extension $1 \rightarrow \mathbb{Z}_2 \rightarrow \hat{\Lambda}_{24} \rightarrow \Lambda_{24} \rightarrow 0$

add cr.-an. op. $\downarrow$

$$V_{\Lambda_{24}}$$

$\mathbb{Z}_2$ -orbifold $\left\{ \begin{array}{l} \text{add } \mathbb{Z}_2\text{-twisted sector } \downarrow \\ V_{\Lambda_{24}} \oplus (V_{\Lambda_{24}})_{\text{tw}} \\ \text{project onto } \mathbb{Z}_2\text{-inv. states } \downarrow \end{array} \right.$

$$V^h$$

$$\begin{array}{ccc} (V_L)^{\mathbb{Z}_2\text{-inv.}} \oplus (V_L)^{\mathbb{Z}_2\text{-inv.}}_{\text{tw}} & & \\ \parallel & & \parallel \\ \textcircled{C} V_0 & & T_0 \textcircled{C} \\ \oplus & & \oplus \\ V_3 & \longleftrightarrow & T_3 \\ & \sigma & \text{triality operator} \end{array}$$

$$\text{Aut}(\Lambda_{24}) = C_0$$

$\text{Hom}(L, \mathbb{Z}_2)$
S11
 $(\mathbb{Z}_2)^{24}$

$$\text{Aut}(\hat{\Lambda}_{24}) = 2^{24} \cdot C_0$$

$\uparrow$
non-split

$$\text{Aut}(V_{\Lambda_{24}}) = 2^{24} \cdot C_0$$

$$\hookrightarrow 2^{1+24} \cdot C_0$$

$$\hookrightarrow 2^{1+24} \cdot C_1$$

$\uparrow$
This action does not mix
the untw. sector and the tw. sector.

Thm (Frenkel, Lepowsky, Meurman) "Vertex operator algebras and the Moonshine" (1988)

$$\text{Aut}(V^h) = \langle 2^{1+24} \cdot C_1, \sigma \rangle \cong M.$$

3. Conway (super)moonshine

3.1. Duncan's module

We will construct an $N=1$ VOSA V^{fg} of $C=12$ and $V^{\text{fg}}|_{\text{weight } \frac{1}{2}} = 0$

Thm (Duncan, 2007)

"Conway supermoonshine module"
"Duncan's module"

$$\text{Aut}_{N=1}(V^{\text{fg}}) \cong C_0,$$

Thm (Duncan, Mack-Crane, 2014)

The uniqueness of V^{fg} is still a conj.
($C=24, V^{\text{fg}}|_{\frac{1}{2}}=0$) (FLM uniqueness conj.)

Such a "VOSA" of $C=12$ without weight- $\frac{1}{2}$ states is unique
a self-dual C_2 -cofinite
rational VOSA of CFT type

up to isom.

• n real chiral free fermions

NS sector (Clifford module VOSA)

As a vec. sp.,

$$F_{NS} = \text{Span}_{\mathbb{C}} \left\{ \underbrace{\psi_{-r_1}^{i_1} \dots \psi_{-r_\ell}^{i_\ell}}_{\text{weight } r_1 + \dots + r_\ell} | 0 \rangle \mid \begin{array}{l} 1 \leq i_j \leq n, \ell = 0, 1, 2, \dots \\ r_j \in (\mathbb{Z} + \frac{1}{2})_{>0}, \{\psi_r^i, \psi_s^j\} = -2\delta^{ij} \delta_{r+s,0} \\ \{\psi_0^i, \psi_0^j\} = 0 \end{array} \right\}.$$

R sector (canonically-twisted module)

$$\Psi_0^i := \frac{1}{\sqrt{2}} (\psi_0^{2i-1} + \sqrt{-1} \psi_0^{2i}), \quad \bar{\Psi}_0^i := \frac{1}{\sqrt{2}} (\psi_0^{2i-1} - \sqrt{-1} \psi_0^{2i}) \quad \leftarrow \text{choice of the polarization}$$

As a vec. sp.,

$$F_R = \text{Span}_{\mathbb{C}} \left\{ \underbrace{\psi_{-r_1}^{i_1} \dots \psi_{-r_\ell}^{i_\ell} \Psi_0^{i'_1} \dots \bar{\Psi}_0^{i'_\ell}}_{\text{weight } r_1 + \dots + r_\ell + \frac{n}{16}} | 0 \rangle_R \mid \begin{array}{l} 1 \leq i_j, i'_j \leq n, \ell, \ell' = 0, 1, 2, \dots \\ r_j \in \mathbb{Z}_{>0}, \{\psi_r^i, \psi_s^j\} = -2\delta^{ij} \delta_{r+s,0} \end{array} \right\}.$$

• Spin(n) action

$\psi^1, \dots, \psi^n$: an orthonormal $\mathbb{C}$ -basis of $V \cong \mathbb{C}^n$

$$\begin{aligned} (\psi^i + \psi^j) \otimes (\psi^i + \psi^j) &= -(\psi^i)^2 - (\psi^j)^2 - 2\langle \psi^i, \psi^j \rangle \\ \psi^i \otimes \psi^i + \psi^j \otimes \psi^j &= -2\delta^{ij} \end{aligned}$$

$$\text{Cliff}(V) := \bigoplus_{i=0}^{\infty} V^{\otimes i} / v \otimes v \sim -|v|^2 \quad \{\psi^i, \psi^j\} = -2\delta^{ij}$$

$$\text{Spin}(n) \subset \text{Cliff}(V)$$

$$1 \rightarrow \mathbb{Z}_2 \rightarrow \text{Spin}(n) \rightarrow \text{SO}(n) \rightarrow 1 \quad \text{a double cover of } \text{SO}(n)$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ \pm \hat{M} & \mapsto & M \end{array}$$

$$\hat{M}(\psi^1 \dots \psi^n) \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} \hat{M}^{-1} = (\psi^1 \dots \psi^n) M \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}$$

$\chi \in \text{Spin}(n)$ acts on

$$F_{NS} \text{ as } \psi_{-r_1}^{i_1} \dots \psi_{-r_\ell}^{i_\ell} |0\rangle \mapsto (\chi \psi_{-r_1}^{i_1} \chi^{-1})_{-r_1} \dots (\chi \psi_{-r_\ell}^{i_\ell} \chi^{-1})_{-r_\ell} |0\rangle \quad \text{SO}(n)\text{-action}$$

$$F_R \text{ as } \psi_{-r_1}^{i_1} \dots \psi_{-r_\ell}^{i_\ell} \mathbb{I}_0^{i_1} \dots \mathbb{I}_0^{i_\ell} |0\rangle_R \mapsto (\chi \psi_{-r_1}^{i_1} \chi^{-1})_{-r_1} \dots (\chi \mathbb{I}_0^{i_\ell} \chi^{-1})_0 \chi |0\rangle_R$$

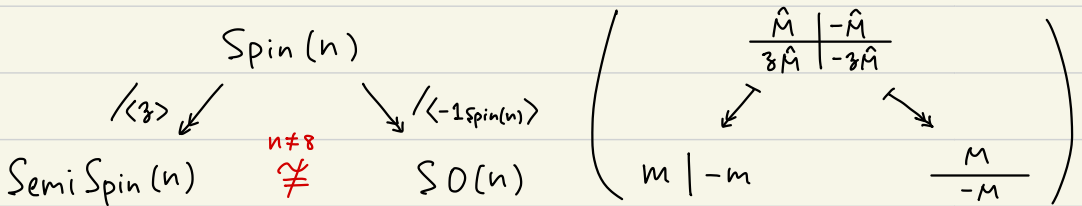
$$\begin{array}{c} \uparrow \\ \text{Cliff}(V) \\ \text{SU} \\ \langle \psi_{-r_1}^{i_1}, \dots, \psi_{-r_\ell}^{i_\ell} \rangle \neq \langle \psi_0^{i_1}, \dots, \psi_0^{i_\ell} \rangle \end{array}$$

When n is even, $-1_{\text{SO}(n)} \in \text{SO}(n)$.

The lifts of $-1_{\text{SO}(n)}$ are $\pm \psi^1 \dots \psi^n =: \pm \mathcal{Z}$.

$$\text{Center}(\text{Spin}(n)) = \begin{cases} \mathbb{Z}_2 = \langle -1_{\text{Spin}(n)} \rangle & (n: \text{odd}) \\ \mathbb{Z}_4 = \langle \mathcal{Z} \rangle & (n \equiv 2 \pmod{4}) \\ \mathbb{Z}_2 \times \mathbb{Z}_2 = \langle -1_{\text{Spin}(n)} \rangle \times \langle \mathcal{Z} \rangle & (n \equiv 0 \pmod{4}) \end{cases}$$

If $n \equiv 0 \pmod{4}$,



parities of each sector:

fermion parity γ	$-1 \text{Spin}(n)$		
	+1	-1	
+1	F_{NS}^0	F_R^0	$n=24$: V^{fh}_{NS} (NS) $\leftarrow \text{SemiSpin}(24)$
-1	F_{NS}^1	F_R^1	$= V^{fh}_{tw}$ (R)
	$\parallel$	$\parallel$	
	F_{NS}	F_R	
	$\hookrightarrow$		
	$\text{SO}(n)$		

Annotations for the mapping:

- " $\langle \gamma \rangle$ -orbifold"
- " $\langle -1 \text{Spin}(n) \rangle$ -orbifold"

minimum-weight states of each sector:

$F_{NS}^0 \ni 0\rangle$	1-dim. wt. 0	$F_R^0 \ni (\text{even } \# \text{ of } \mathbb{Z}_2 \text{'s}) 0\rangle_R$	2"-dim. wt. $\frac{3}{2}$	positive chiral rep. of $\text{Spin}(24)$
$F_{NS}^1 \ni \psi_{-\frac{1}{2}}^i 0\rangle$	24-dim. wt. $\frac{1}{2}$	$F_R^1 \ni (\text{odd } \text{---}) 0\rangle_R$	2"-dim. wt. $\frac{3}{2}$	negative ---

No weight $-\frac{1}{2}$ states in V^{fh} .

• $C_{0,}$ action on V^{fq}

Fact

$$\Downarrow \quad \equiv C_{0,} \subset Spin(24)$$

$\swarrow \quad \searrow \quad \parallel 2$

$$SemiSpin(24) \supset C_{0,} \\ \downarrow \\ V^{fq}$$

$$C_{0,} = Aut(\Lambda_{24}) \subset SO(24)$$

Fact $det = +1$ $\nearrow$

In the 2^{11} -dim. weight $-\frac{3}{2}$ subsp. $(V^{fq})_{\frac{3}{2}}$,

there exists a 1-dim. inv. subsp. under the $C_{0,}$ action.

Its basis G constitutes the supercurrent of the $C_{0,}$ -inv. $N=1$ SCA.

$$\rightarrow Aut_{N=1}(V^{fq}) \cong C_{0,}.$$

Stolz-Teichner? proposal? program?

3.2. Connection to the Segal-Stolz-Teichner conjecture

Conj cf. [1108.0189 Stolz-Teichner]

$$SQFT \cong TMF.$$

↑ the space ($\mathbb{Z}$ -graded ring)
of the classes of topological modular forms

↑ the space ($\mathbb{Z}$ -graded ring)
of the equiv. classes of 2d $N=(0,1)$ SQFTs

For $[\tau] \in SQFT_n$, degree

first
Pontryagin class

· the anomaly polynomial of the grav. anom. = $\frac{n}{48} p_1$

· $SQFT_n \xrightarrow{\sim} TMF_n \rightarrow (MF_{\mathbb{Z}}^{w-h})_{\frac{n}{2}}$ weakly holomorphic modular forms
with $\mathbb{Z}$ coefficient of weight $\frac{n}{2}$

$$f\left(\frac{a\tau+b}{c\tau+d}\right) = (c\tau+d)^{\frac{k}{2}} f(\tau)$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ [\tau] & \xrightarrow{\quad\quad\quad} & \eta(\tau)^n Z_{ell}\tau \end{array}$$

$$Z_{ell}\tau := \text{Tr}_{\mathcal{H}_{\tau, R\text{-sector}}} (-1)^F q^{\frac{H+P}{2}} \bar{q}^{\frac{H-P}{2}}$$

If τ is an SCFT,

· $n = 2(c_R - c_L)$

· $Z_{ell}\tau = \text{Tr}_{\mathcal{H}_{\tau, R\text{-sector}}} (-1)^{F_L + F_R} q^{L_0 - \frac{c_L}{24}} \bar{q}^{\tilde{L}_0 - \frac{c_R}{24}}$

↑
indep. of $\bar{q}$
(the function of q)

If we assume that the SST conj. is true,

we can obtain nontrivial statements on SQFT.

by translating the properties of TMF.

· Example 1. the divisibility property

$TMF_n \rightarrow (MF_{\mathbb{Z}}^{w-h})_{\frac{n}{2}}$ is not surjective.

Its image is also determined. (Hopkins, Mahowald) cf. [Hopkins Math/0212397 Prop. 4.6]

In particular,

$$(\text{its image}) \cap \mathbb{Z}[\eta(\tau)] = \begin{cases} \mathbb{Z} \left[\frac{24}{\gcd(24, \frac{n}{24})} \eta(\tau)^n \right] & (n \equiv 0 \pmod{24}) \\ 0 & (\text{otherwise}) \end{cases}$$

Therefore, if the SST conj. is true, then for $[\tau] \in \text{SQFT}_n$,

$$\mathbb{Z}_{\text{ell}}\tau \text{ is const.} \Rightarrow \begin{cases} \frac{24}{\gcd(24, \frac{n}{24})} \mid \mathbb{Z}_{\text{ell}}\tau & (24 \mid n) \\ \mathbb{Z}_{\text{ell}}\tau = 0 & (\text{otherwise}) \end{cases}$$

For example,

$$V^{f_4} : (c_L, c_R) = (12, 0) \quad n = -24$$

$$\frac{24}{\gcd(24, \frac{n}{24})} = 24 \quad \mid \quad \mathbb{Z}_{\text{ell}}[V^{f_4}](\tau) = -24$$

• Example 2. the periodicity element

Fact

There exists a periodicity element $X \in \mathrm{TMF}_{-24^2} = -576$ s.t.

$$X \cdot : \mathrm{TMF}_n \xrightarrow{\sim} \mathrm{TMF}_{n-576}.$$

Under the SST conj., this translates to the existence of

an $N=(0,1)$ SQFT τ of $N=-576$ s.t. $Z_{\mathrm{ell}}[\tau] = 1$.

It suffices to find

an $N=1$ chiral SCFT τ of $C=288$ s.t. $Z_{\mathrm{ell}}[\tau] = 1$.

$$\downarrow$$

$$Z_{\mathrm{ell}}\tau = \text{const.}$$

$(\mathrm{VfH})^{\otimes 24}$ has $C = 24 \times 12 = 288$.

[1909.08388 Johnson-Freyd]

The anomaly of its C_0 sym. is described by the generator of $\mathrm{SH}(C_0) \cong \mathbb{Z}_{24}$.

$\leadsto (\mathrm{VfH})^{\otimes 24} / C_0$ has $Z_{\mathrm{ell}}[\tau] = 1$?

$(\mathrm{VfH})^{\otimes 24} / (C_0 \times A_{24})$ $\neq$? [2210.14923 Albert, Lin, Kaidi]

$C_0 \times S_{24}$ is anomalous.

4. Mathieu moonshine

cpt. Kähler mfd. with triv. can. bdl.

K3 surface : a cpx-2-dim. CY mfd. with $h^{0,1} = 0$.

$$\begin{smallmatrix} 0 & 1 \\ 0 & 20 & 0 \\ 1 & 0 & 0 \end{smallmatrix}$$

K3-target SCFT : $\mathcal{N} = (4, 4)$, $(c_L, c_R) = (6, 6)$

The elliptic genus the weak Jacobi form of weight 0 and index $\frac{c_L}{6} = 1$

$$Z_{K3}(\tau, z) := \text{Tr}_{\mathcal{H}_{K3}} (-1)^{F_L + F_R} y^{J_0} q^{L_0 - \frac{c_L}{24}} \bar{q}^{\tilde{L}_0 - \frac{c_R}{24}} \quad \left(\begin{matrix} \tau = e^{2\pi i \tau} \\ y = e^{2\pi i z} \end{matrix} \right)$$

$$= 2\mathcal{Y}_{0,1}(\tau, z)$$

$$= 2 \{ y + 10 + y^{-1} + (10y^2 - 64y + 108 - 64y^{-1} + 10y^{-2})y + \dots \}$$

If we decompose Z_{K3} into the irr. char. of the $\mathcal{N}=4$ SCA of $c=6$,

$$Z_{K3}(\tau, z) = 20 \text{ch}_{\frac{1}{4}, 0}^{(\text{massless})}(\tau, z) - 2 \text{ch}_{\frac{1}{4}, \frac{1}{2}}^{(\text{massless})}(\tau, z)$$

$$+ 2 \sum_{n=1}^{\infty} A_n \text{ch}_{n+\frac{1}{4}, \frac{1}{2}}(\tau, z).$$

n	1	2	3	4	5	6	...
A_n	<u>45</u>	<u>231</u>	<u>770</u>	<u>2277</u>	<u>5796</u>	13915	...
						<u>3520</u> + <u>10395</u>	

dim. of irrep. of M_{24} :

1, 23, 45, 45, 231, 231, 252, 253, 483, 770, 770,

970, 970, 1035, 1035, 1035, 1265, 1771, 2024,

2277, 3312, 3520, 5313, 5796, 5544, 10395

→ underlying VOA-like object?

K3 Mathieu moonshine (三上, 大栗, 立川, 2010)

other evidence of the existence of VOA

- twined ell. genera (the McKay-Thompson series) ^{Cheng (2010)}
Gaberdiel, Hohenegger, Volpato (2010)
 $\exists M_{24}$ -module reproducing them. Gannon (2012)
- twisted twined ell. genera

Their mod. trsf. are controlled by a 3-cocycle in $H^3(M_{24}; U(1))$.

(fit into the theory of anomaly) ^{Gaberdiel, Persson, (2012)}
Ronellenfisch, Volpato

However, ^{preserve the hol. 2-form}

- $\text{Aut}_w(K3) \subsetneq M_{23}$ ^{Mukai (1988)}
- $M_{24} \not\subset \text{Aut}_{N=(4,4)}(K3 \text{ CFT}) \subsetneq \text{Co}_1$ ^{Gaberdiel, Hohenegger, Volpato (2010)}

The underlying VOA-like object is not yet found!

solution?

- symmetry surfing? ^{Taormina, Wendland (2011-13)}
- reflected $K3 = V^{fg}$? ^{Duncan, Mack-Crane (2015)}
^{Taormina, Wendland (2019)}
- preserve only $N=(4,1)$? ^{Harvey, Moore (2020)}
some subalgebra?
- non-inv. sym.? ^{Volpato (2024)}
^{Anzies, Giaccari, Harrison, Volpato (2025)}

⋮